

E9 205 Machine Learning for Signal Processing

Linear Models for Regression and Classification

11-10-2017

Linear Regression

$$y(\mathbf{x}, \mathbf{w}) = \sum_{j=0}^{M-1} w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x})$$

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon$$

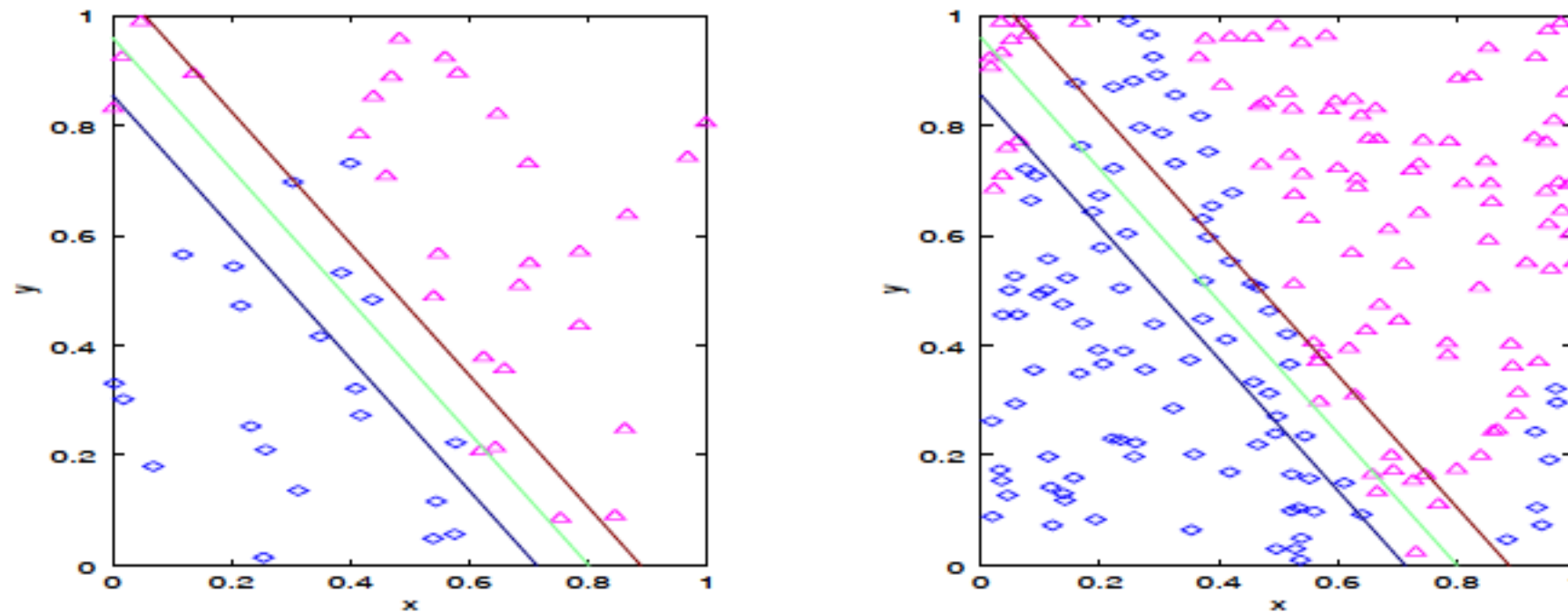
- ❖ Solution to Maximum Likelihood problem is the least squares solution

$$\nabla \ln p(\mathbf{t}|\mathbf{w}, \beta) = \sum_{n=1}^N \{t_n - \mathbf{w}^T \phi(\mathbf{x}_n)\} \phi(\mathbf{x}_n)^T.$$

Pseudo Inverse Based Solution

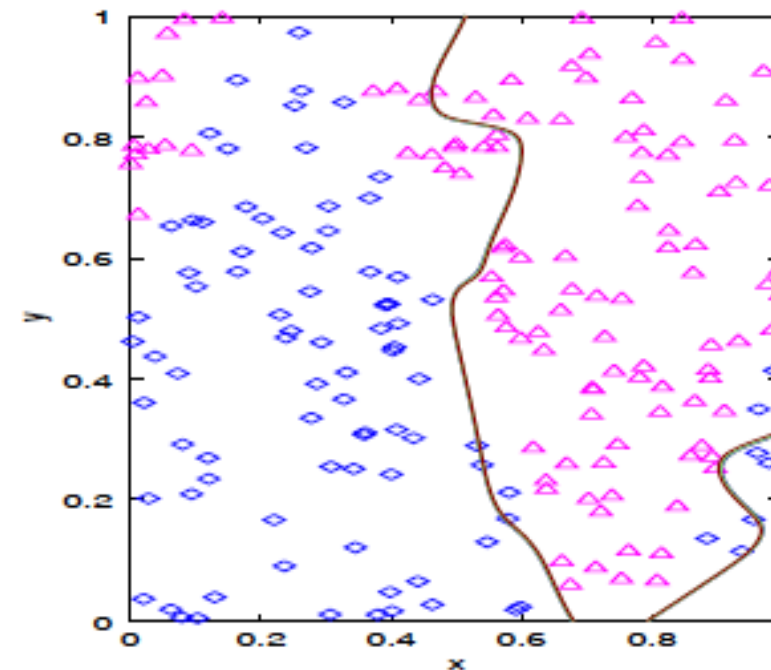
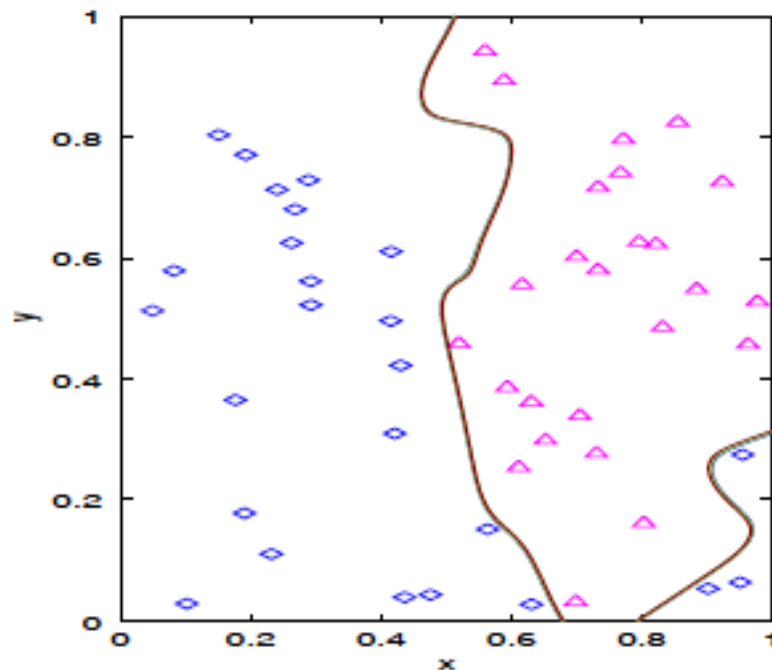
Bishop - PRML book (Chap 3)

Underfit



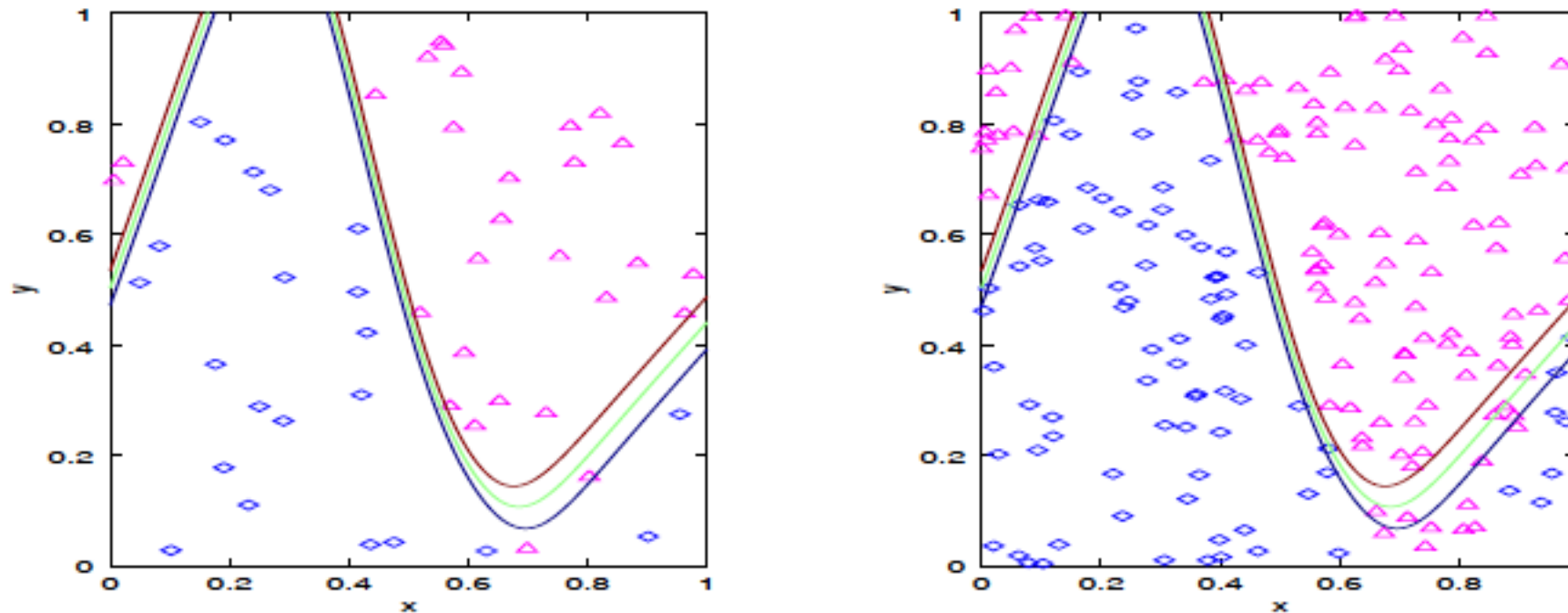
- The model is not able to capture the variability in the data (Linear Model)
- Both the training and testing error are high (15%,20%)
- Try to learn a more complex model – more features, more hidden neurons, decrease regularization
- More data would not help

Overfit



- The model is capturing data as well as accidental variations (100 hidden neurons)
- Training error is too low and testing error is too high (0%, and 16%)
- Try to learn a simpler model – less features, less hidden neurons, increase regularization
- More data would help

Compromise

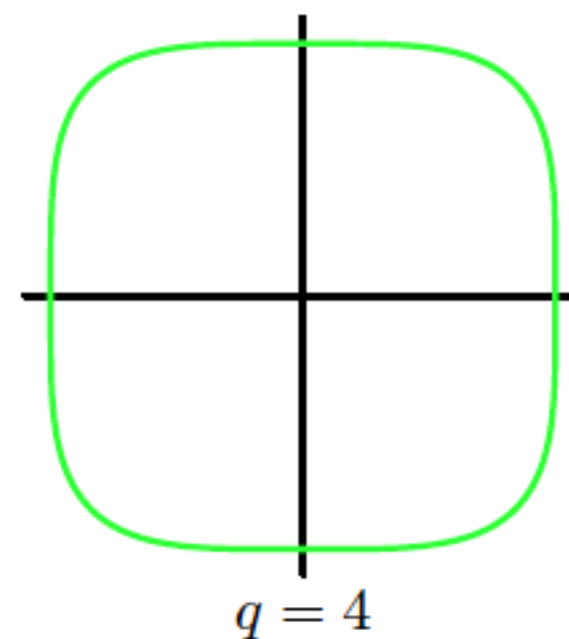
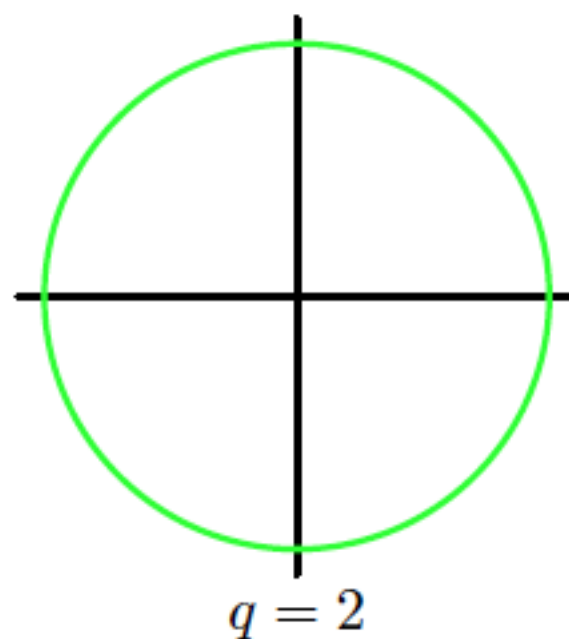
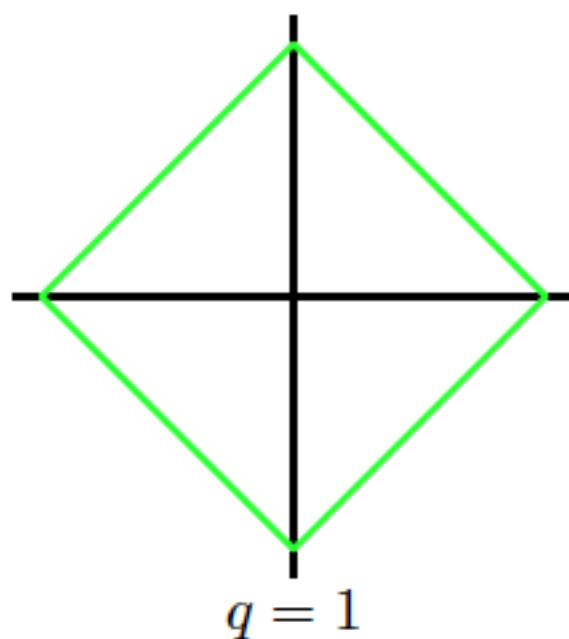
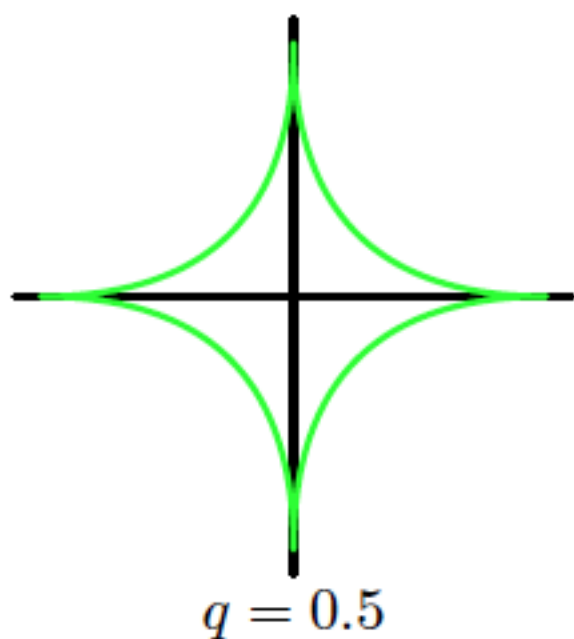


- Reasonable training and test errors – (4%, 8%)
- Appropriate model – capturing only the global characteristics not details

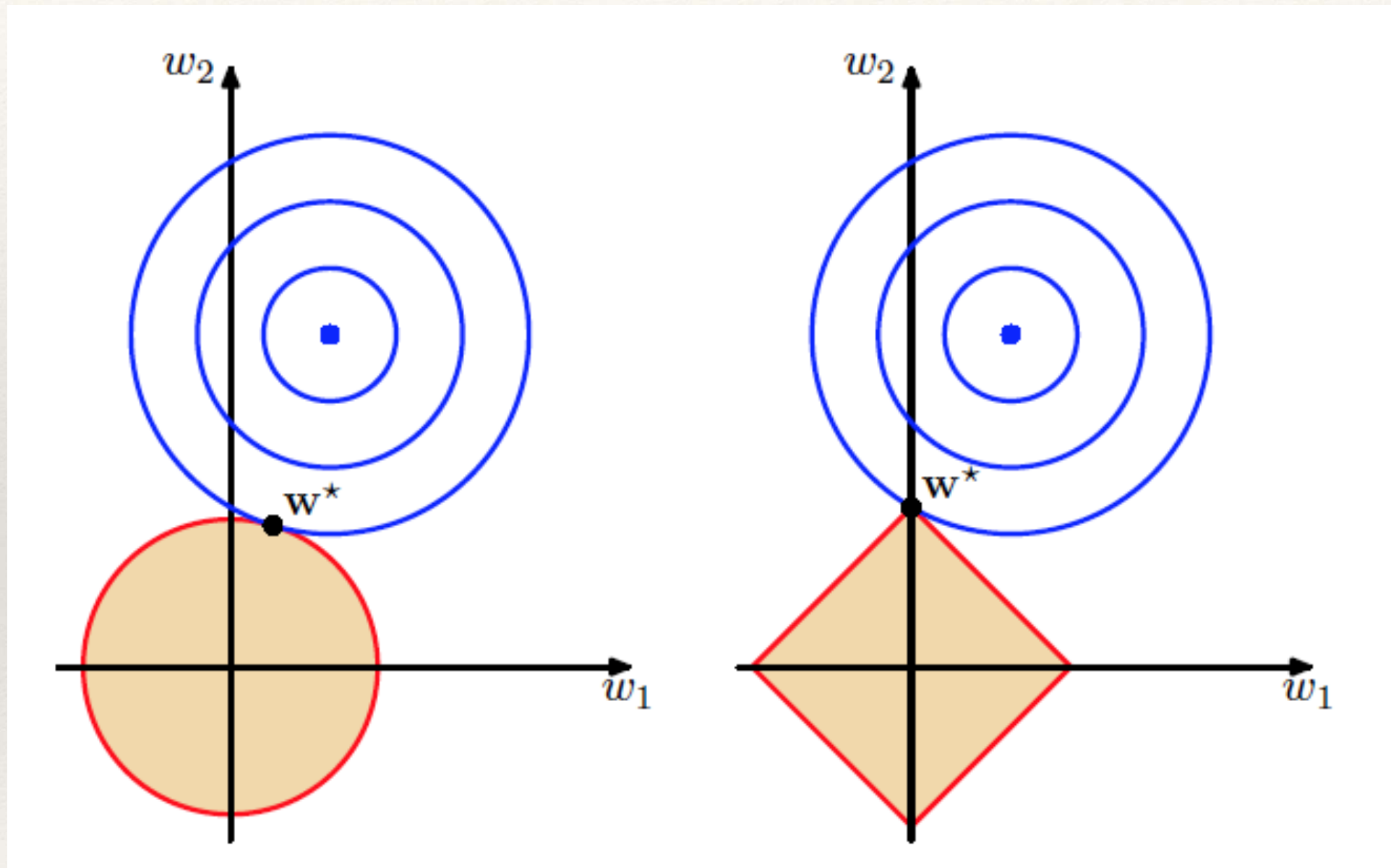
Regularized Least Squares

- ❖ Optimize a modified cost function

$$E_D(\mathbf{w}) + \lambda E_W(\mathbf{w})$$



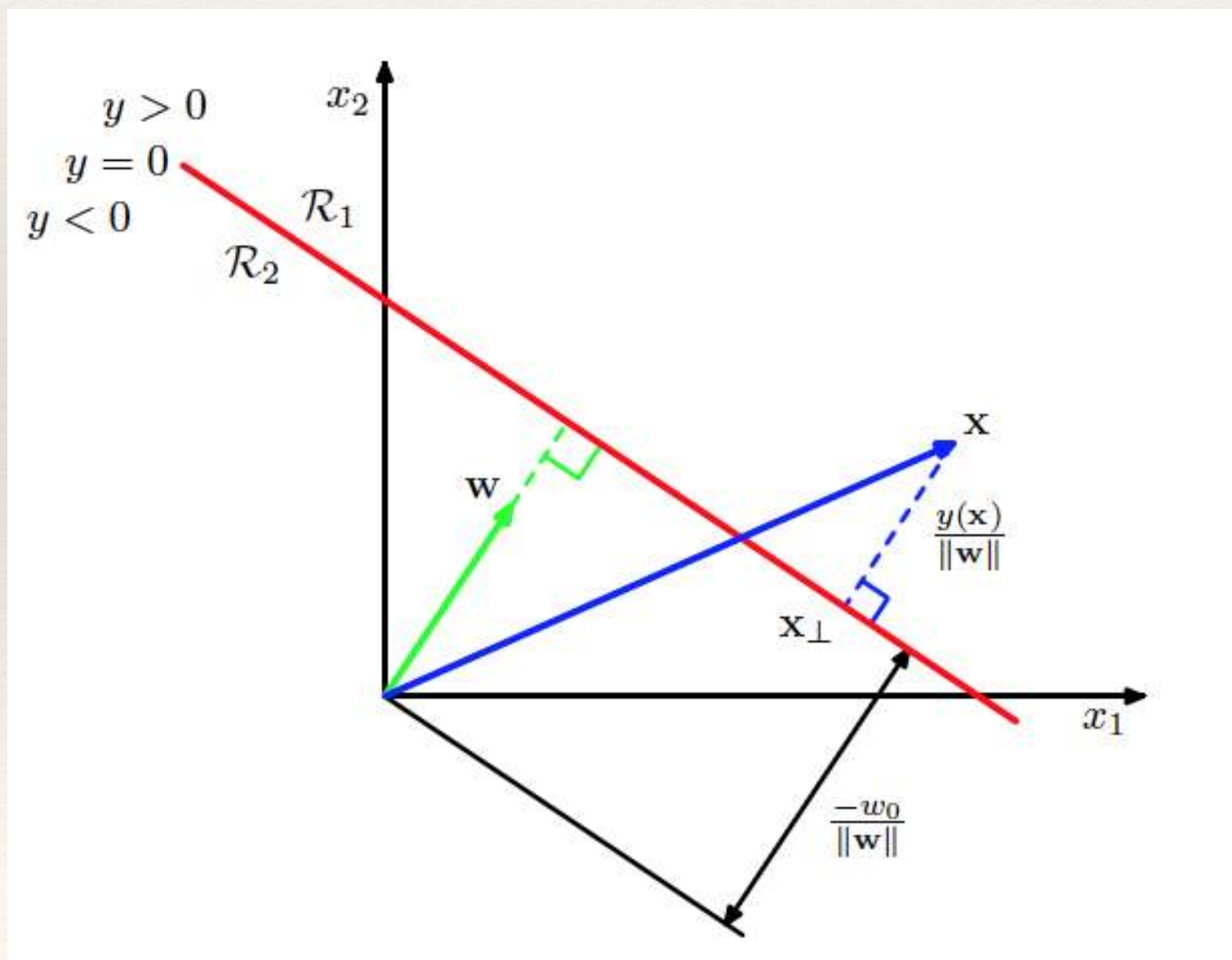
Regularized Least Squares



Linear Models for Classification

- ❖ Optimize a modified cost function

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$$



Least Squares for Classification

- ❖ K-class classification problem

$$y_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

$$\mathbf{y}(\mathbf{x}) = \widetilde{\mathbf{W}}^T \widetilde{\mathbf{x}}$$

- ❖ With 1-of-K hot encoding, and least squares regression

$$E_D(\widetilde{\mathbf{W}}) = \frac{1}{2} \text{Tr} \left\{ (\widetilde{\mathbf{X}} \widetilde{\mathbf{W}} - \mathbf{T})^T (\widetilde{\mathbf{X}} \widetilde{\mathbf{W}} - \mathbf{T}) \right\}$$

Logistic Regression

- ❖ 2- class logistic regression

$$p(C_1|\phi) = y(\phi) = \sigma(\mathbf{w}^T \phi)$$

- ❖ Maximum likelihood solution

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N (y_n - t_n) \phi_n$$

- ❖ K-class logistic regression

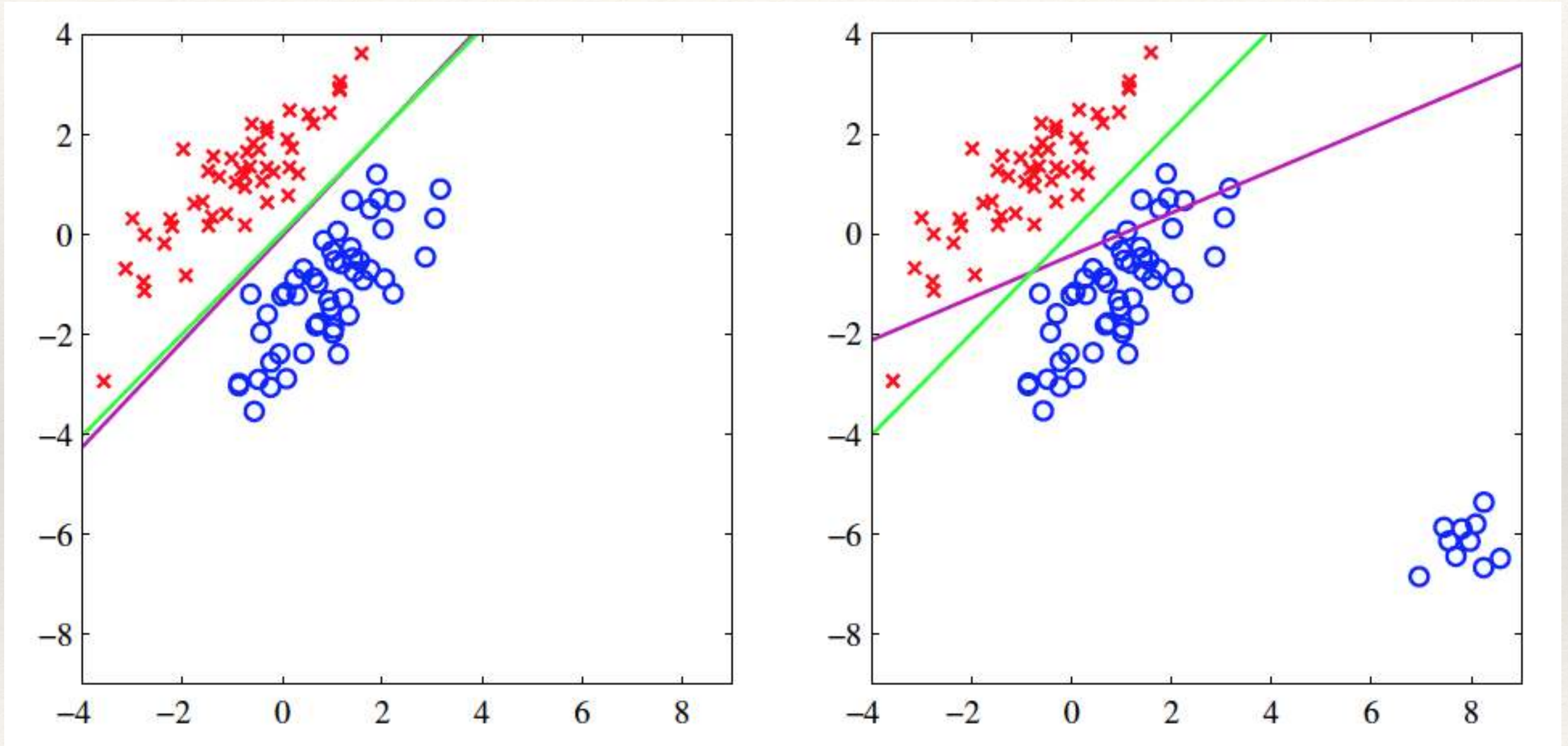
$$p(C_k|\phi) = y_k(\phi) = \frac{\exp(a_k)}{\sum_j \exp(a_j)}$$

- ❖ Maximum likelihood solution

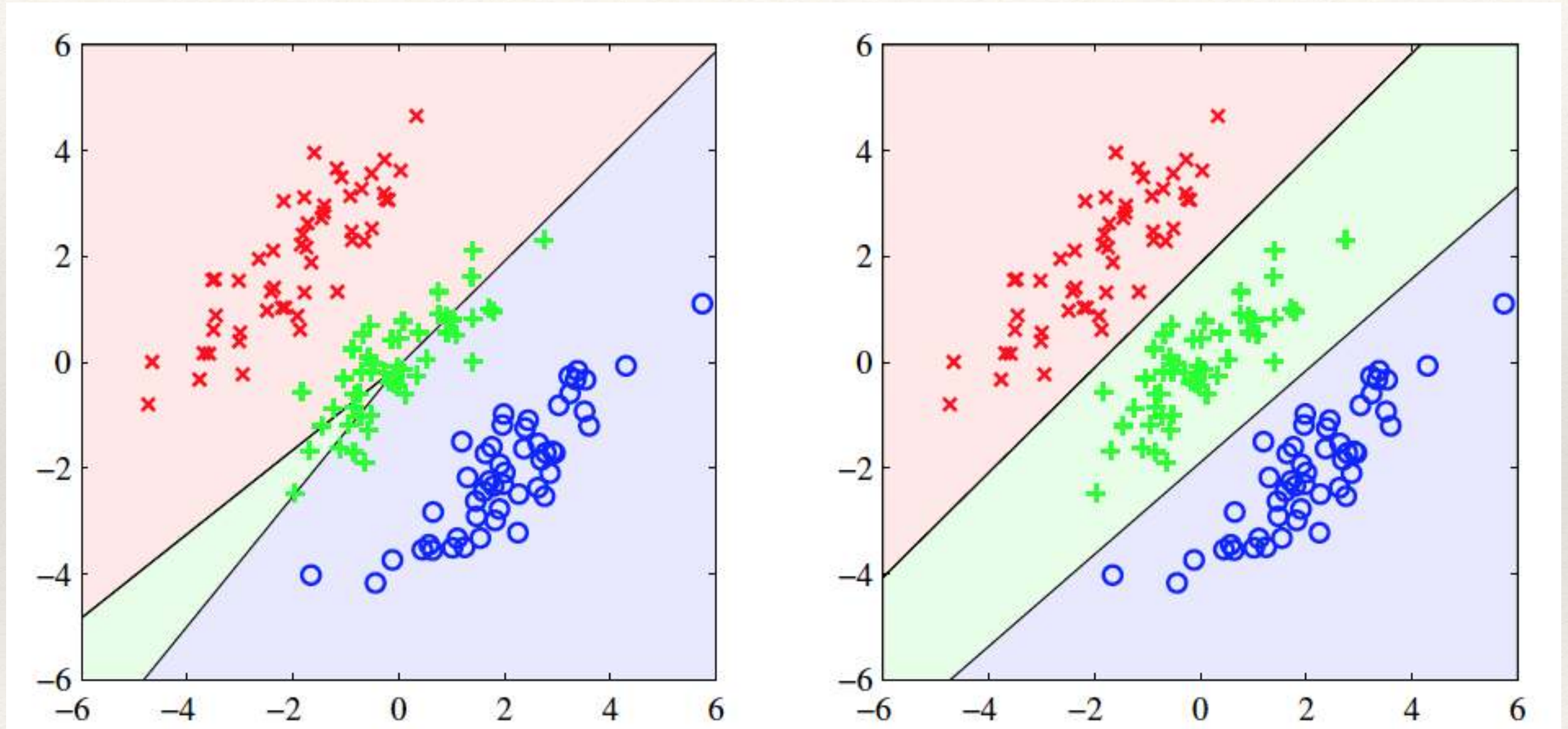
$$a_k = \mathbf{w}_k^T \phi.$$

$$\nabla_{\mathbf{w}_j} E(\mathbf{w}_1, \dots, \mathbf{w}_K) = \sum_{n=1}^N (y_{nj} - t_{nj}) \phi_n$$

Least Squares versus Logistic Regression



Least Squares versus Logistic Regression



Bishop - PRML book (Chap 4)